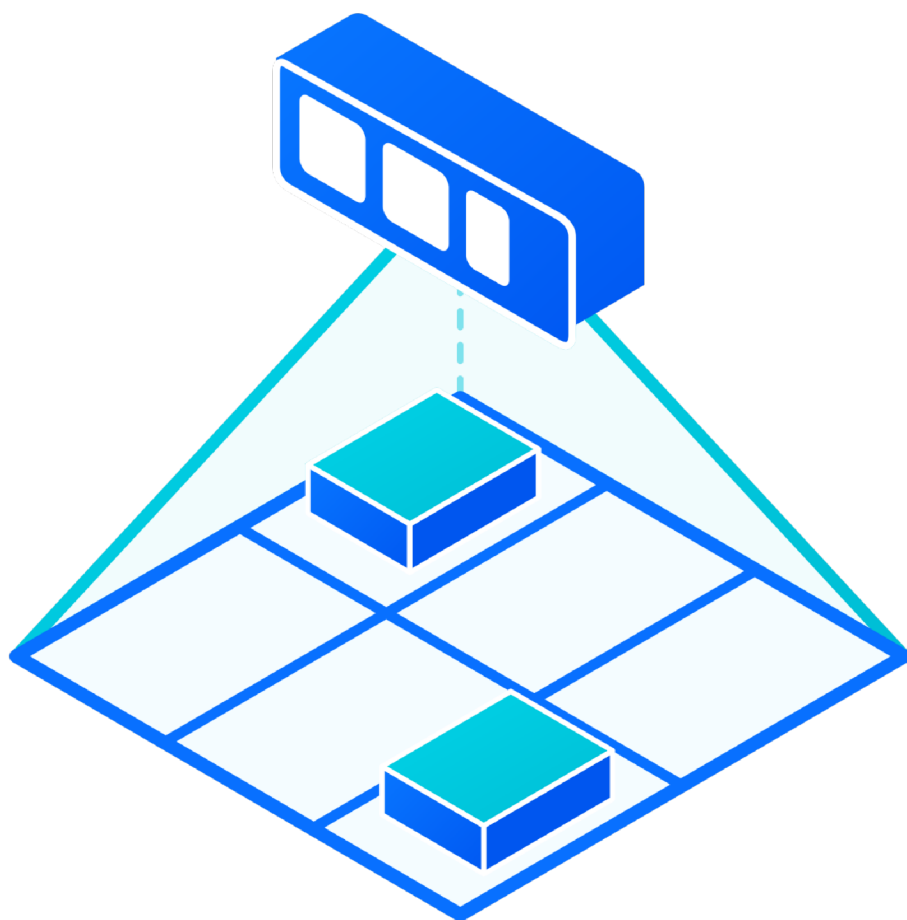




AW3 StockSync

Storage Slot Detection White Paper

Detect slot occupancy and placement for inspections, inventory checks and checks before picking or placing cargo.

Zhejiang MRDVS Technology Co., Ltd.

V0.1 · 2026-09-24

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1 Product Overview

StockSync runs on the AW3 algorithm platform. It detects whether each storage slot is empty or occupied and provides placement quality, dimensions and coverage when enabled. The standard S10 Pro RGB-D camera supports slot inspections, inventory checks and checks before picking or placing cargo.

1.1 Applications



Slot inspections

Check each slot
Find empty and occupied slots



Inventory checks

Group by camera and slot ID
Send results to business systems



Before picking or placing

Check presence, bounds, alignment
Support picking and placing

Slot inspections, inventory checks and checks before picking or placing cargo

Warehouse application



Floor storage: occupancy by slot supports inspections, inventory checks and cargo handling

Image source: MRDVS warehouse case study

Scope The solution provides slot detection and AW3 configuration tools. Inventory records, scheduling and cargo handling remain with the customer's or a third-party business system.

1.2 Standard Camera: S10 Pro

The standard S10 Pro RGB-D camera supplies RGB images and depth data. Camera specifications and algorithm operating limits are described separately.



S10 Pro (left) and coverage orientation (right)

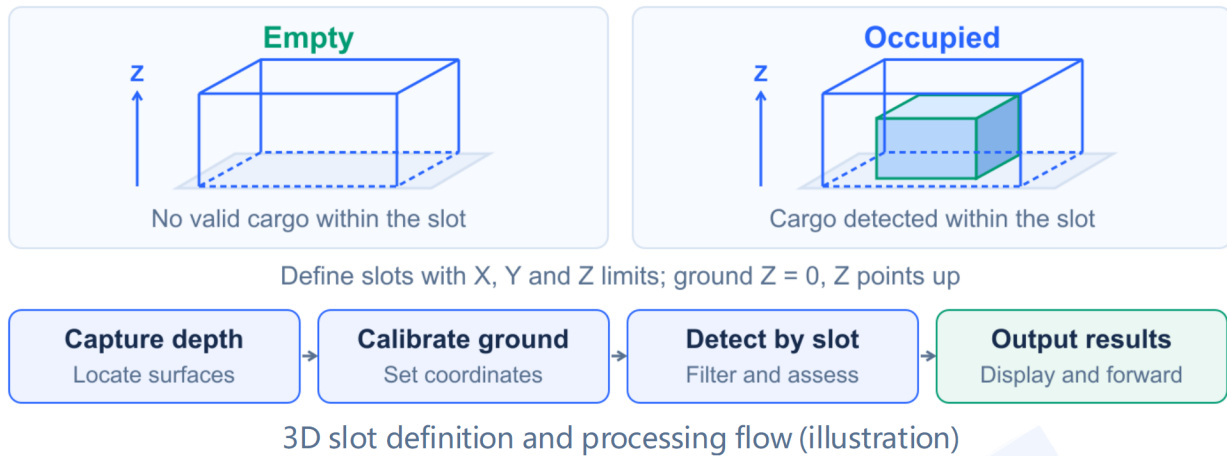
Key specification	S10 Pro
Output data	Depth, RGB and IR amplitude images
Depth resolution / frame rate	240 × 160; typical and maximum 10 fps
RGB resolution / frame rate	1920 × 1080; typical and maximum 10 fps
Field of view H × V	TOF and RGB: 61° × 90° ± 3°
Camera range	0.3–17 m (90% reflectivity) 0.3–13 m (10% reflectivity)
Ranging accuracy	≤ 3 cm
Power / average consumption	24 V; ≤ 7 W
Communication / time sync	Ethernet, IO; PTP
Dimensions / weight	92 × 47 × 51 mm; 460 g
Protection / temperature	IP67; –20°C to 60°C
Laser / eye safety	940 nm; CLASS I

Mounting orientation The S10 Pro has a wider field of view along the short side of its housing. Do not copy the mounting orientation of other MRDVS models. Align this side with the direction requiring wider coverage and confirm it in the AW3 live view. Generic camera symbols do not indicate mounting orientation. Camera range and accuracy do not define the algorithm's mounting height, dimensional accuracy or detection success rate. See Section 3.2 for algorithm limits.

Source: MRDVS S10 Series RGBD Camera Product Specification, V2.1.

2 How It Works

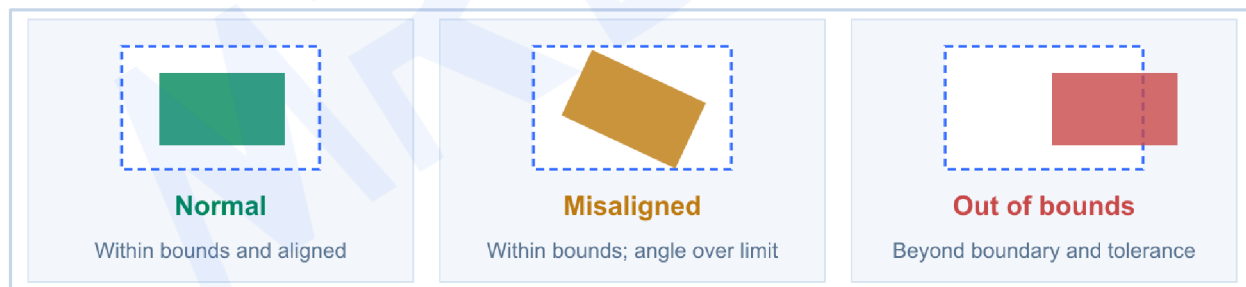
2.1 From Depth Data to Slot Results



The camera captures depth data. The algorithm checks occupancy within each calibrated slot and, when enabled, evaluates placement and object dimensions. Ground and noise filtering reduce background interference.

2.2 Occupancy and Placement Quality

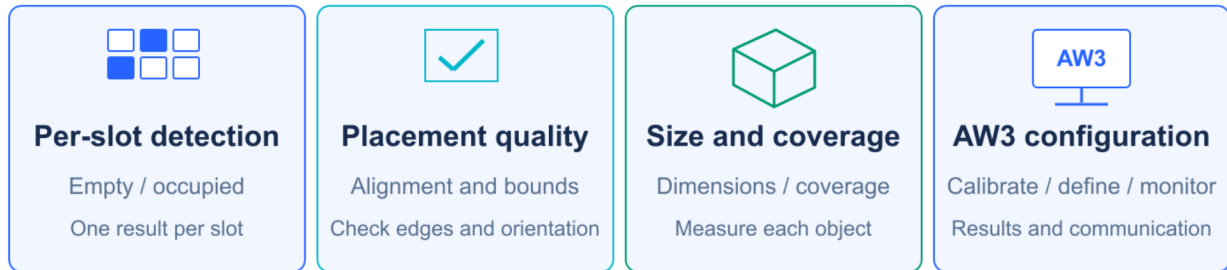
Check	Meaning
Empty / occupied	Empty: no valid cargo in the slot. Occupied: cargo detected.
Placement quality	When enabled, checks normal, misaligned and out-of-bounds states. Angles are not checked when out of bounds.



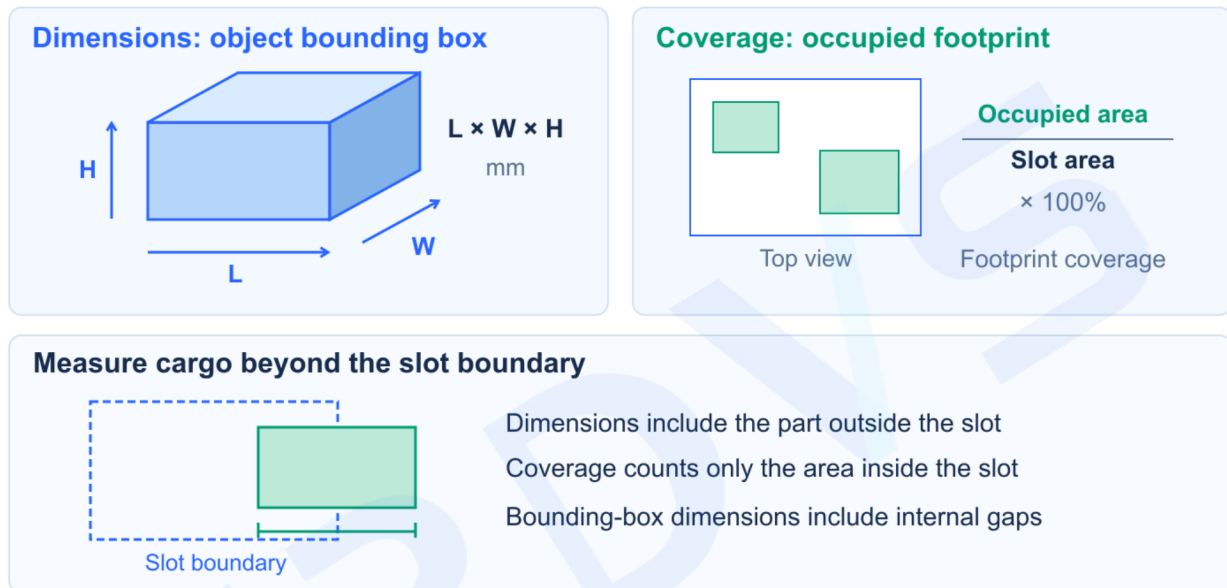
Criteria Out of bounds means the cargo's bounding rectangle exceeds the allowed boundary. Misaligned means it is within bounds but exceeds the angle limit. Angle checks apply to rectangular or palletized cargo with stable straight edges.

3 Product Benefits

3.1 Features and Outputs



Occupancy, placement and geometry, configured together in AW3



Dimensions and coverage (illustration)

Feature	Value
Separate occupancy and placement	Check both cargo presence and whether it is out of bounds or misaligned.
Per-object dimensions	Report each object's length, width and height, including any part outside the slot.
Count and coverage	Report object count and footprint coverage by slot to help assess occupancy.
Batch slot setup	Reduce repeated setup with batch slot creation, shared limits and a preview of all regions.

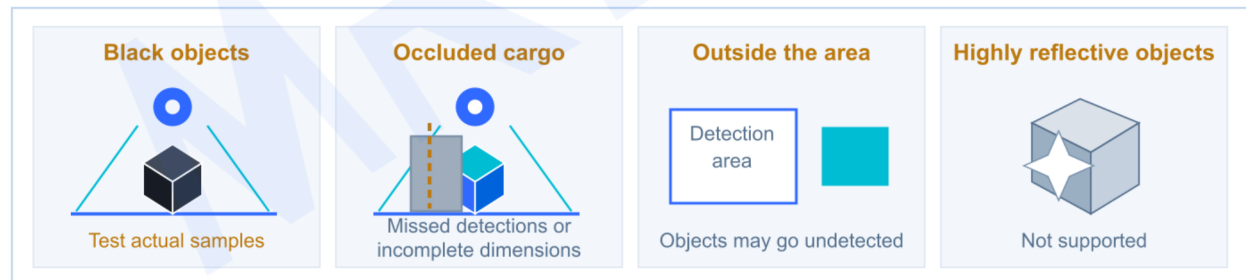
Solution scope The core functions are slot occupancy and placement checks. Dimensions and coverage help assess space usage. For comprehensive cargo dimensioning or multi-view reconstruction, consider a dedicated volume measurement solution.

3.2 Performance and Operating Limits

The table lists operating conditions and dimensional targets that require validation.

Item	Range / criterion	Conditions
Mounting height: occupancy	≤8 m	Recommended; keep all cargo in view.
Mounting height: size and placement	Target limit: 5 m; unverified	Measurement of 40 cm objects at 5 m is not yet validated.
Object size for dimensions	Each dimension ≥400 mm; validate at the installed height	Verify minimum object sizes for both measurement and occupancy detection.
Absolute dimension error	Length, width, height each ≤4 cm	Acceptance target; verify each dimension at the agreed height using representative samples.
Material	Test black objects on site; highly reflective objects are unsupported	Very dark or low-reflectivity objects may be missed or yield no measurements.
Coverage and slot count	Depends on height, field of view and cargo size	Multiple slots per camera; cameras detect independently.
Environment	Low dust, no rain; normal indoor/outdoor light	Fix camera securely; clean lens and revalidate after changes.

Factors that affect detection



Material, occlusion and detection range affect results

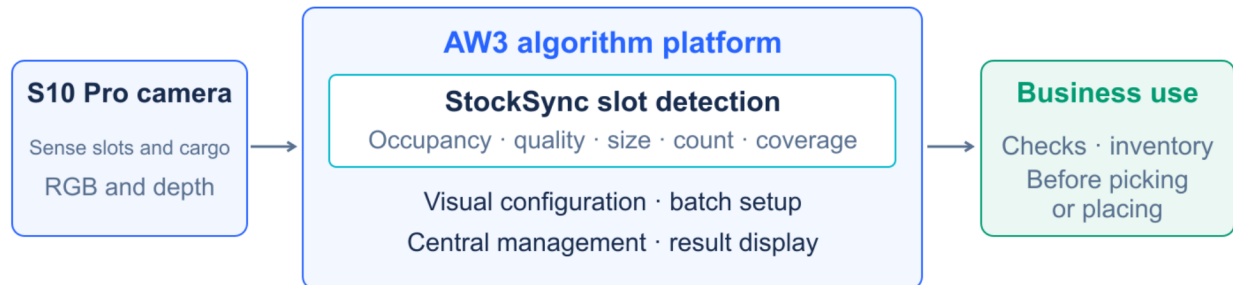
Result definitions Dimensions describe the object's bounding box, including gaps and any part outside the slot. Coverage is the percentage of the slot's footprint occupied by cargo.

Site validation At the actual mounting height and lighting, test detection and dimensional accuracy separately using representative cargo, including very dark samples. Revalidate after changes to object size, material or site conditions.

4 Deployment

4.1 Central Management in AW3

Install StockSync on the camera via SSH. Use AW3 to connect to the camera, configure slots and view results.



The camera captures the scene; AW3 manages detection and supplies results to business applications

4.2 Deployment Steps



Step	Task
Connect and inspect	Install the camera; keep slots and cargo in view.
Ground calibration	Establish the ground reference and spatial coordinates.
Configure slots	Define slots; use batch creation for regular layouts.
Verify and enable	Choose functions and test empty and occupied cases.

4.3 Site Configuration

Parameters filter out ground and interference, balance response time and stability, and enable dimension or placement outputs. Adjust them to suit the cargo and operating cycle.

Deployment Visual setup, batch slot creation and shared boundary settings reduce repeated work. Deployment time depends on mounting conditions, slot count and verification scope.

5 Implementation and Use

5.1 Implementation and Acceptance



Confirm suitability, then deploy and verify on site

Agree on cargo, slot layout, mounting and response requirements before deployment. After deployment, test representative and boundary samples for accuracy, repeatability, stability and response time. Keep configuration and acceptance records.

Test	Expected result
Occupancy	Stable empty/occupied results matched to the correct slot.
Enabled functions	Placement, dimensions and coverage meet business needs.
Continuous use	Response time and sustained operation meet the required operating cycle.

5.2 Operating Requirements

Item	Requirement
Mounting and view	Secure camera; cover all slots; keep lens clean and avoid persistent occlusion.
Cargo and environment	Use within the agreed size, material and environmental limits. See Section 3.2.
Site changes	Recalibrate and verify after camera, support surface or layout changes.

Project assessment Outside the recommended conditions, assess the site and cargo samples before agreeing on the solution and delivery requirements.

6 Frequently Asked Questions

Question	Answer
Where can it be used?	Slot inspections, inventory checks and checks before picking or placing cargo, providing occupancy data to business systems.
What results are available?	Empty/occupied detection, with optional placement quality, dimensions, count and footprint coverage.
Are black or reflective objects supported?	Test actual samples. Very dark or low-reflectivity cargo may be missed or yield no measurements. Highly reflective cargo is unsupported. Detection alone does not establish dimensional accuracy.
Does occupied mean correctly placed?	No. Occupancy and placement are separate. Placement checks identify out-of-bounds and misaligned cargo.
Is coverage the same as volume utilization?	No. Coverage is the percentage of the slot's footprint occupied by cargo. Dimensions describe each object's bounding box, including its out-of-bounds portion.
Can it identify cargo classes?	Semantic recognition is not standard in this version. Contact the supplier for supported classes and the required setup.
How can coverage be expanded?	Consider additional cameras based on your slot layout. Cameras detect independently and are managed in AW3; adding cameras does not automatically stitch views.
Must existing inventory or scheduling systems be replaced?	No. Slot detection supplies scene data. Existing systems retain inventory, scheduling and cargo handling.
Is deployment straightforward?	AW3 provides visual setup, batch slot creation and shared limits. Time depends on site installation and verification needs.

Consultation Provide the slot layout, mounting height, representative cargo and required outputs so the supplier can assess suitability.

Revision History and Support

Version	Date	Changes
V0.1	2026-09-24	Product overview, principles, benefits, deployment and operating requirements.

Technical Support

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